Bayesian Regression for Real Estate Pricing in Buenos Aires

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Bayesian Regression for Real Estate Pricing

Introduction

In this project, real estate prices in Buenos Aires are analyzed using Bayesian linear regression. The goal is to build models that predict the price of a listing using its property characteristics. We compare 3 likelihood assumptions using PSIS-LOO cross-validation that reveals which model provides the best predictive performance.

1. Normal likelihood
2. Student T likelihood
3. Outlier Classifier (mixture model)

Data

The data was gathered from a public dataset containing real estate listings in Buenos Aires.

Source: Kaggle data "Buenos Aires real estate"

Number of observations (after cleaning): 14,200.

Variables of interest:

- price → Dependent variable. Quantitative. Continuous.
- rooms → Predictor. Quantitative. Discrete.
- surface_total_in_m2 → Predictor. Quantitative. Continuous.

Each variable was standardized (subtracted the mean and divided by the standard deviation) to facilitate easier prior specification.

Model 1: Normal Likelihood

$$y \sim \text{Normal}(\mu_i, \sigma)$$

$$\mu_i = \alpha + \beta_1 * x_1 + \beta_2 * x_2$$

where x_1 and x_2 are standardized rooms and surface area variables, respectively.

$$\alpha \sim \text{Normal}(0,\ 5)$$

Intercept. Before seeing the data, we expect the price of properties to be around the mean when the number of rooms and surface area are at their average values. However, this prior is weakly informative (a high standard deviation of 5) so it will allow for variation.

$$\beta_1 \sim \text{Normal}(0.2, 0.5)$$

Slope for rooms. We believe more rooms generally increase price of a listing. Particularly, we expect a unit increase in standard deviation in rooms to be associated with a 0.2 standard deviation increase in price. Still, flexibility is allowed, meaning the slope could be weak, strong, or even negative if the data suggests it.

$$\beta_2 \sim \text{Normal}(0.3, 0.3)$$

Slope for surface area. We believe surface area has a stronger effect than rooms and are slightly more confident that the effect of surface area is positive, so we give it a narrower standard deviation.

$$\sigma \sim \text{HalfNormal}(1)$$

We expect typical price deviations to be around 1 in standardized units. Using a half-normal to ensure sigma is always positive.

The data was fit to this model and the sampler's performance was assessed. All parameters had $\widehat{R} = 1$, the trace plots didn't show major trends, and the rank plots were approximately uniform, indicating good convergence. The pair plots had single peaks, suggesting the posterior is unimodal (the chain isn't bouncing between separate modes). This means the posterior estimates are reliable. Refer to the Appendix for the supporting plots and tables.

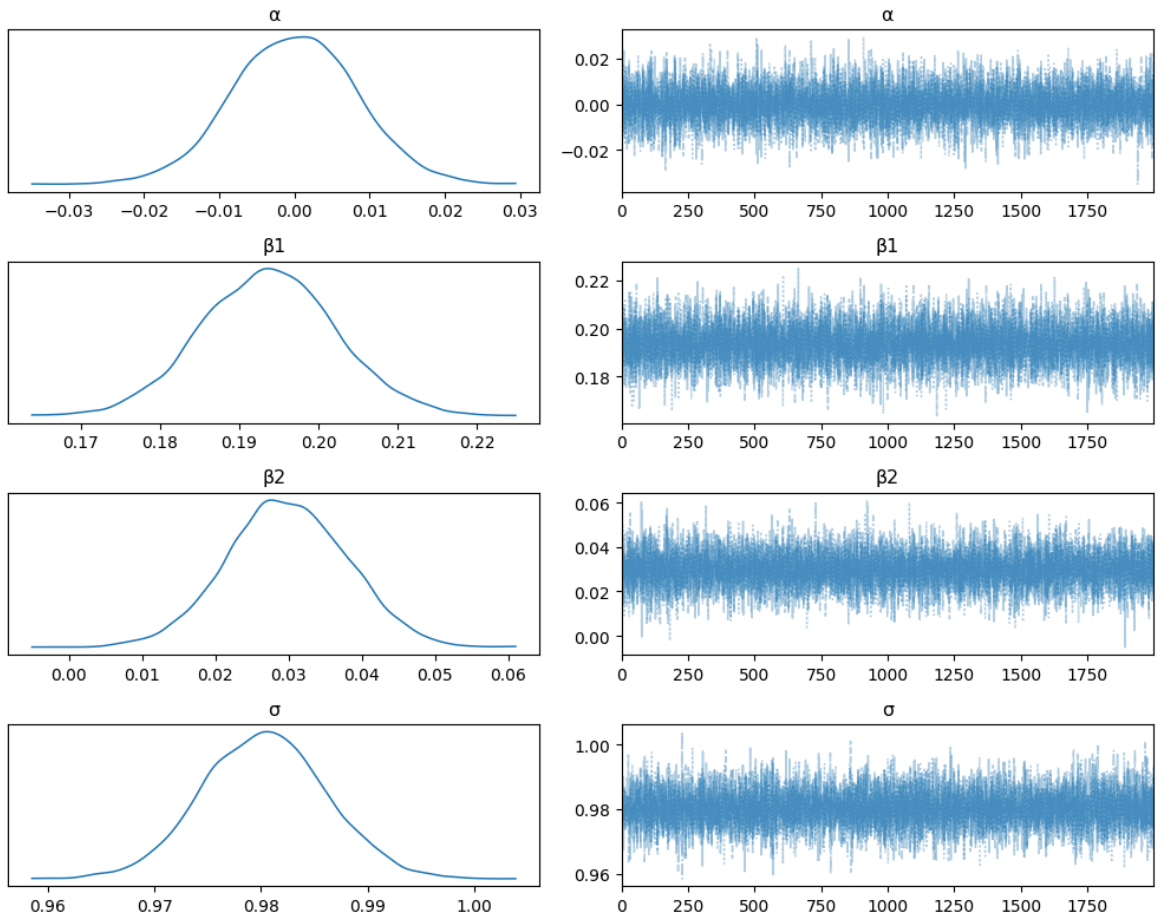


Image 1: Posteriors: Normal model.

This model suggests that the intercept is around 0, which is expected. Despite expectations, number of rooms have more of an effect on price than surface area. Standard deviation is expected to be around 0.98, meaning even after accounting for rooms and surface area, there is still roughly a full standard deviation of unexplained variation in price. The effect of rooms and surface area are relatively small.

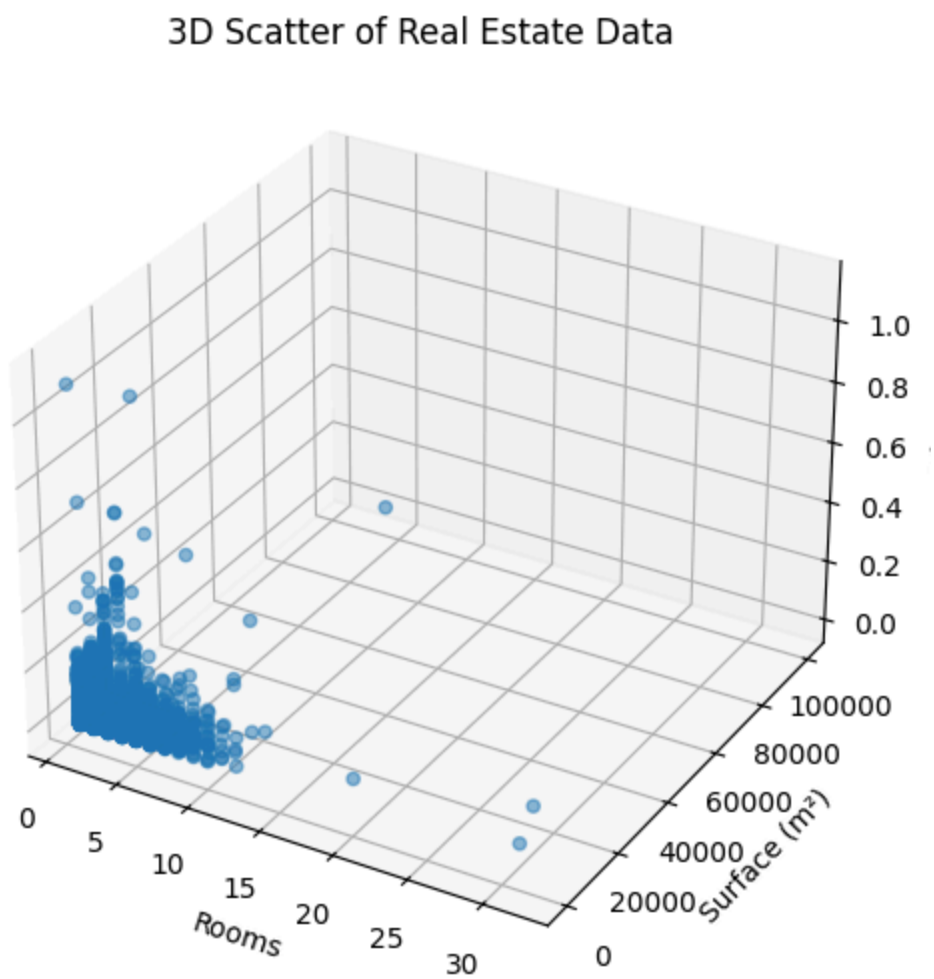


Image 2: 3D Scatter of the data.

It seems that there are a few outliers that may be affecting the predictive power of the normal model. Let's try a more robust model for comparison.

Model 2: Student T Likelihood

$$y \sim \mathrm{T}(\nu, \mu_i, \sigma)$$

$$\mu_i = \alpha + \beta_1 * x_1 + \beta_2 * x_2$$

$$\alpha \sim \mathrm{Normal}(0,\ 5)$$

$$\beta_1 \sim \mathrm{Normal}(0.2, 0.5)$$

$$\beta_2 \sim \mathrm{Normal}(0.3, 0.3)$$

$$\sigma \sim \text{HalfNormal}(1)$$

$$\nu \sim \text{Exponential}(\lambda = 1/30)$$

The new parameter, ν, is the degree of freedom in the Student-T likelihood. The prior assumes that small values of ν are more likely, but its expected value is 30, allowing for a range of degrees of freedom from very small (heavier tails) to large (nearly Normal behavior).

The sampler also worked properly, showing similar metrics as the previous model. Refer to the Appendix for details.

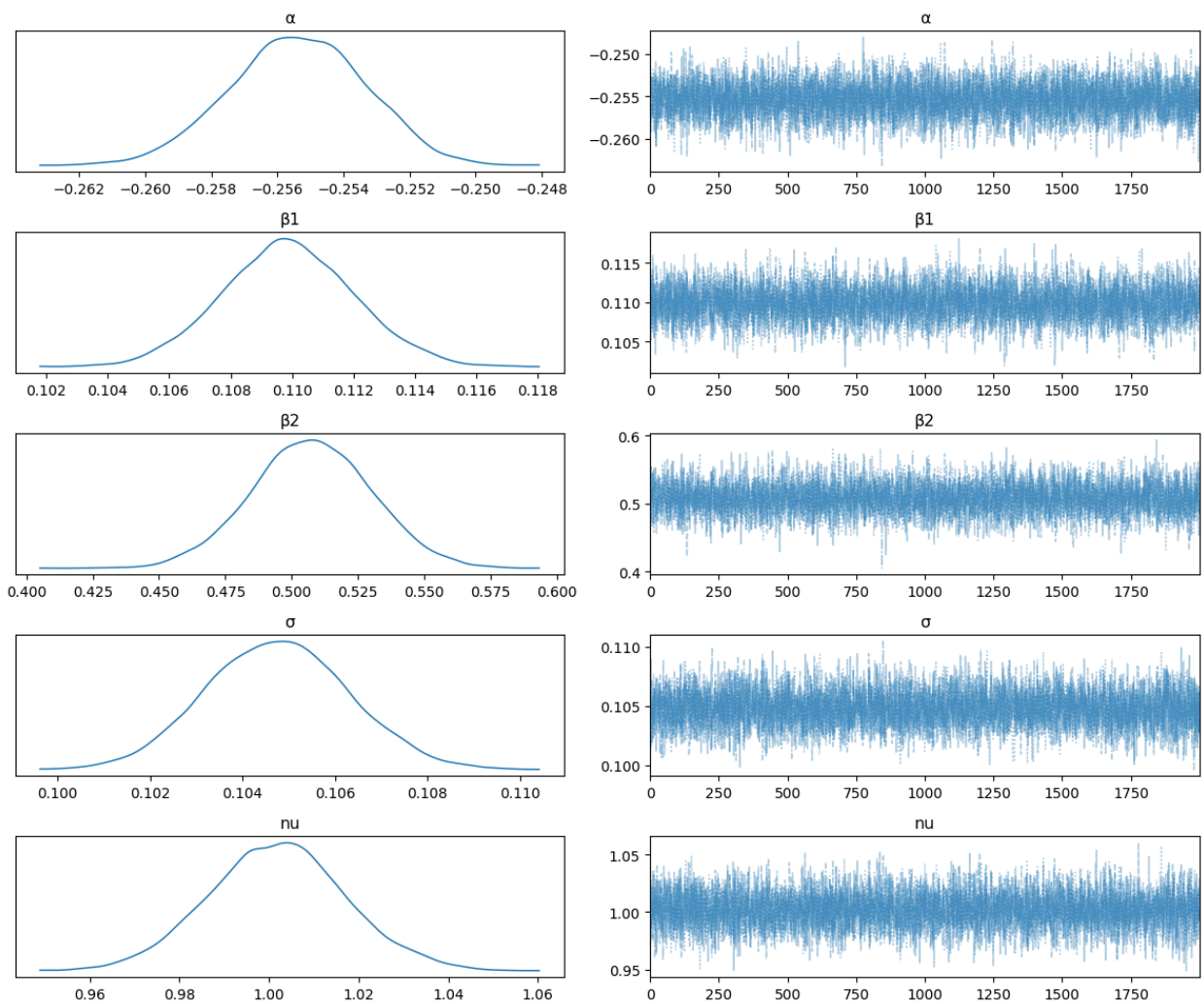


Image 3. Posteriors: Robust model.

The robust model suggests that when the number of rooms and surface area of a property are at their means, the expected price is ~0.25 standard deviations less than the average price. Now that we've introduced the Student-T likelihood, the effect of surface area is much higher and the effect of rooms is slightly lower. ν is expected to be around 1, which means there are some

extreme outliers and the likelihood responded by adjusting its tails to be thicker.

Model 3: Outlier Classifier

Rather than viewing outliers as random deviations, we can hypothesize that these extreme points stem from an entirely different process. For instance, the luxury real estate market may behave very differently from typical properties, leading to prices that do not fit the usual pattern. By assigning a distinct variance to these points, we can better capture their fundamentally different nature and avoid distorting the main regression fit for the majority of observations.

$$y_i \sim \text{Normal}(\mu_i, \sigma_i)$$

$$\mu_i = \alpha + \beta_1 * x_1 + \beta_2 * x_2$$

$$\alpha \sim \text{Normal}(0,\ 5)$$

$$\beta_1 \sim \text{Normal}(0.2, 0.5)$$

$$\beta_2 \sim \text{Normal}(0.3, 0.3)$$

$$\sigma_i = \sigma_{in} \quad \text{if } q_i = 0$$

$$\sigma_i = \sigma_{in} + \sigma_{out} \text{ if } q_i = 1$$

The new variable q_i is an indicator variable that represents whether or not the model thinks the data point is an outlier. It is 1 if it is, and 0 if not.[1]

$$\sigma_{in} \sim \text{HalfNormal}(0.5)$$

$$\sigma_{out} \sim \text{HalfNormal}(2)$$

In this prior, the inlier points have a baseline standard deviation that is much smaller than that of the outliers.

$$p \sim \text{Uniform}(0, ½)$$

This is the probability that any given data point is an outlier. Between 0 and 50% of all data

[1] #variables: I identified the dependent and independent variables that form the foundation of my three models, ensuring a clear distinction between predictors and responses. I also classified the variables as quantitative (continuous or discrete) to determine appropriate modeling techniques. In the third model, I introduced an indicator variable to classify observations as inliers or outliers, allowing the model to account for distinct variance structures. Additionally, I considered the role of parameters that define the system's constraints and influence model behavior.

points could be outliers. For each point i, it will draw a q_i~Bernoulli(p).

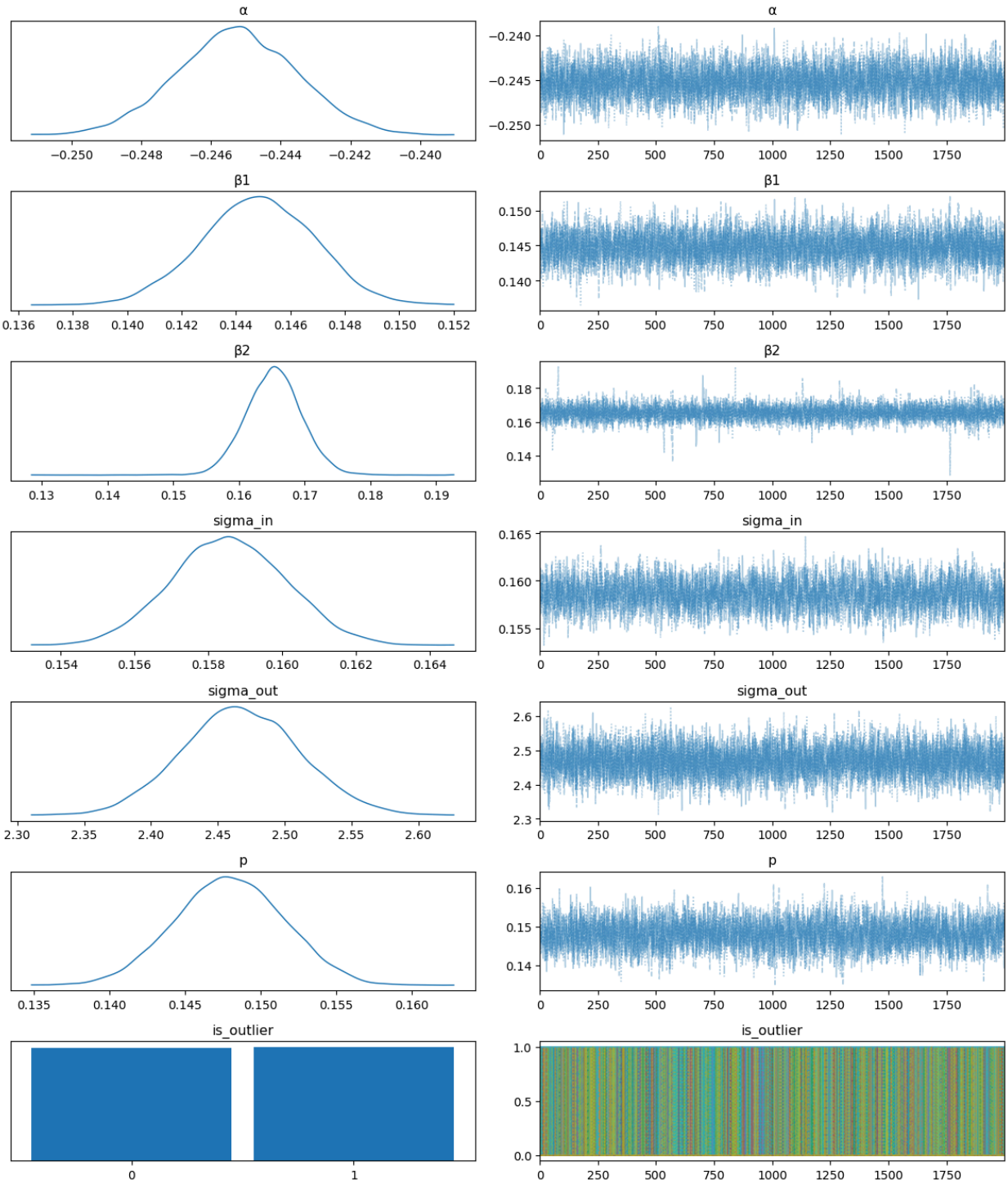


Image 4: Posteriors: Outlier Classifier Model

$\widehat{R}$ values were ones for all parameters except for the last data point's q indicator, for which $\widehat{R}$ was undefined. This likely means all chains agreed the last data point is an outlier (the sampler produced 0 variance in all chains). The rank plots looked uniform, each parameter showed a single peaked distribution, and the chains fluctuate around fairly constant means with no obvious drift. The plots indicate good convergence and effective mixture classification. Refer to the Appendix for the plots.

The classifier model also suggests that the expected price is a bit below average when rooms and surface area are at their average values. The slope of rooms and surface area are both small, around 0.14 and 0.16 standard deviations, respectively. The outliers vary much more than the inliers.

Model Comparison

	rank	elpd_loo	p_loo	elpd_diff	weight	se	dse	warning	scale
Outlier Classifier	0	1701.851629	1386.559989	0.000000	0.999349	355.136640	0.000000	True	deviance
Normal Likelihood	1	7766.633154	10.914917	6064.781525	0.000000	422.902338	234.371137	False	deviance
StudentT Likelihood	2	7766.831681	11.035886	6064.980052	0.000651	422.905734	234.345122	False	deviance

Image 5: PSIS-LOO comparison

Outlier Classifier

- Has the best predictive performance by a large margin.
- The "warning" flag arises because removing certain data points in LOO can radically change how the model assigns them to inlier/outlier groups, making some points highly influential. Despite that, this model's ability to explicitly capture a separate variance for outliers significantly improves estimated predictive accuracy compared to the others.

Normal vs. Student T

- Both produce substantially higher deviance scores, indicating worse predictive accuracy than the Outlier Classifier.
- The robust model (Student T) performed slightly worse than the Normal model, suggesting that a single heavy-tailed distribution was not enough to effectively account for the outliers. This may be due to the outliers being too extreme, causing the likelihood to become overly diffuse. As a result, the model's fit to typical data points deteriorated, leading to a slight decline in overall performance.

In conclusion, rooms and surface area alone do not fully explain price variation, suggesting that additional factors influence real estate pricing. Future models should incorporate categorical variables, such as neighborhood and property type, which may account for much of the remaining variability. While these predictors alone are limited, the outlier classifier model demonstrates some predictive power, and therefore should be favored over the other models considered in this report.

Appendix

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
α	0.00	0.01	-0.02	0.02	0.0	0.0	11984.58	5896.34	1.0
β1	0.19	0.01	0.18	0.21	0.0	0.0	13520.38	6663.61	1.0
β2	0.03	0.01	0.01	0.04	0.0	0.0	14325.34	6395.54	1.0
σ	0.98	0.01	0.97	0.99	0.0	0.0	13779.30	6357.34	1.0

Image 6: Model 1's (Normal) Summary Statistics.

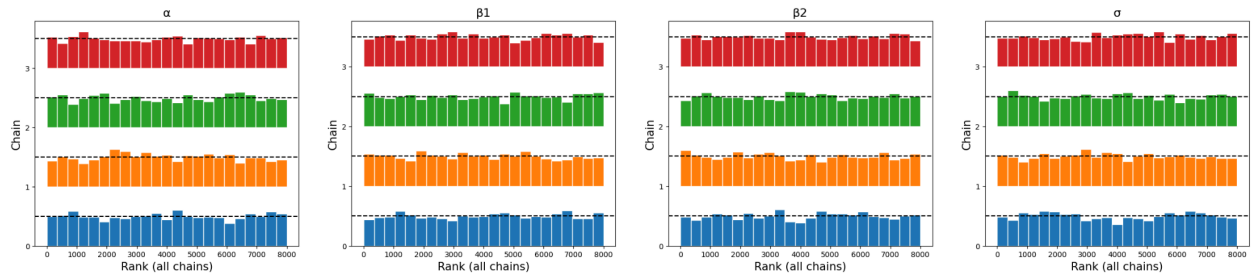


Image 7. Model 1's Rank Plots

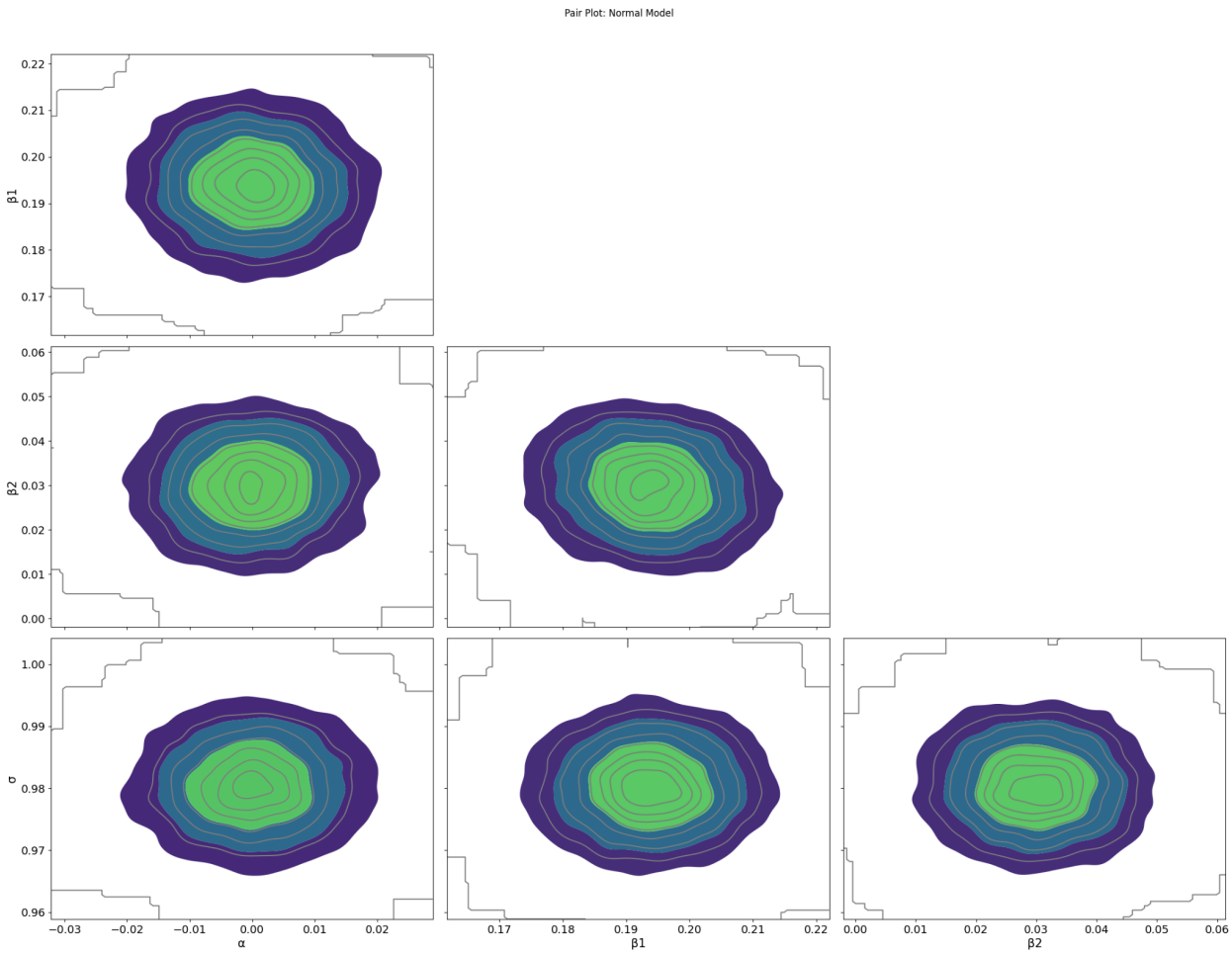


Image 8: Model 1's Pair Plots.

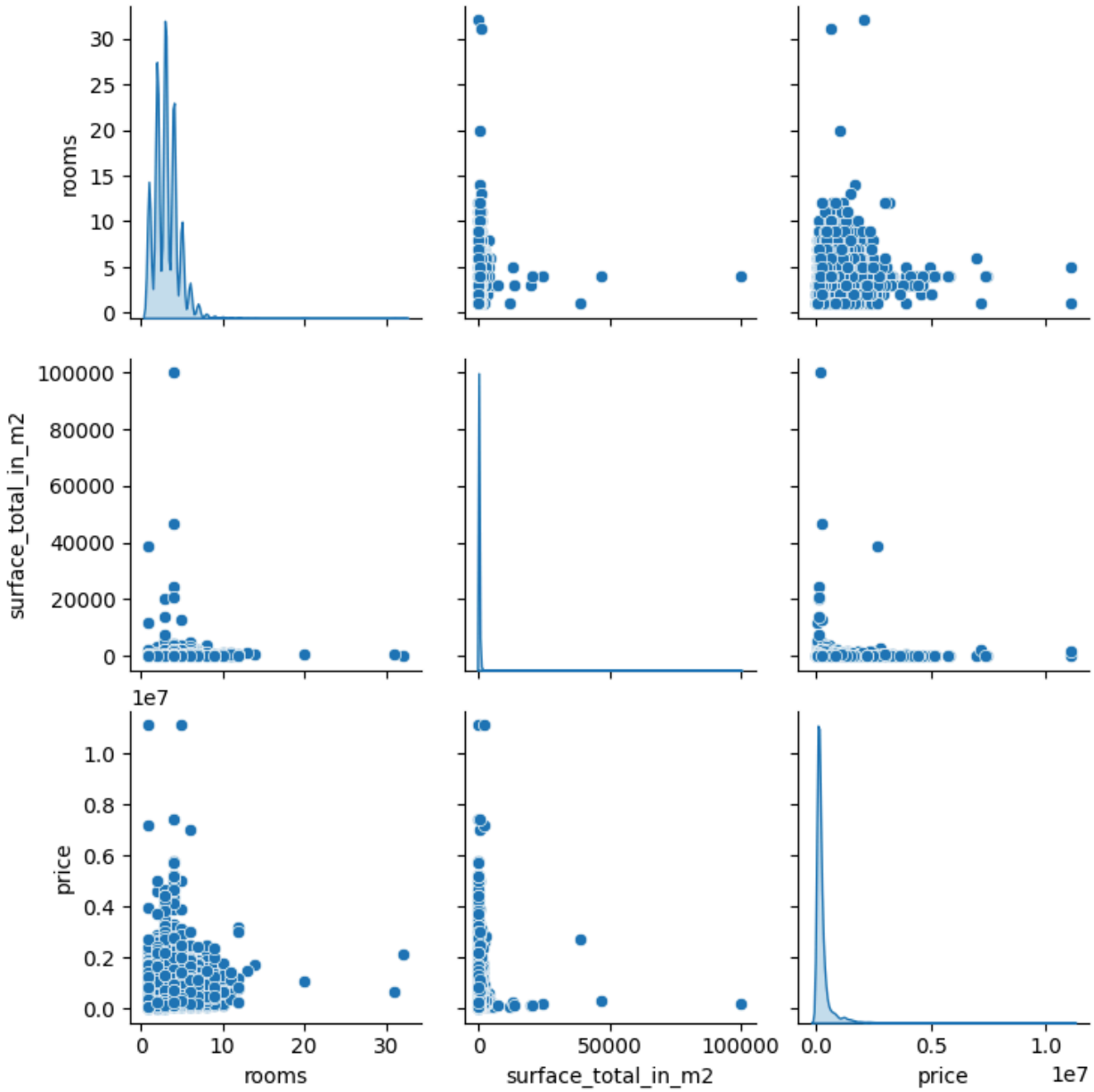
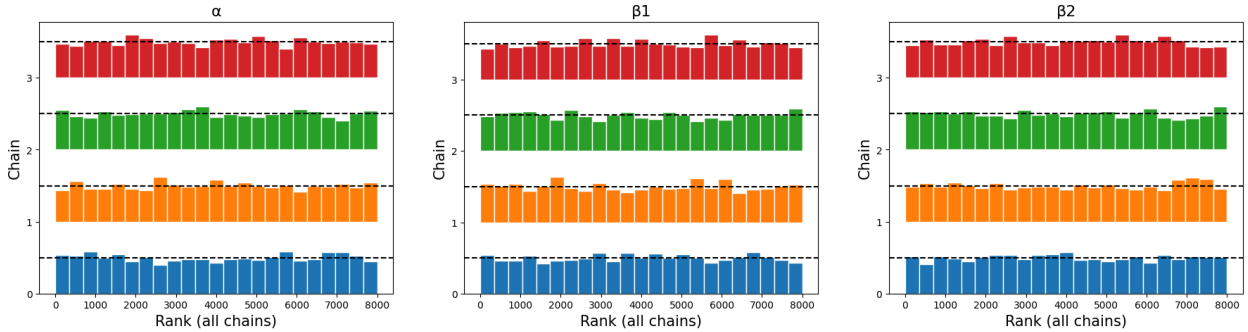


Image 9: Data Pair Plots. Showcases the extreme outliers.

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
α	-0.26	0.00	-0.26	-0.25	0.0	0.0	6850.09	6205.88	1.0
β1	0.11	0.00	0.11	0.11	0.0	0.0	5684.28	5985.14	1.0
β2	0.51	0.02	0.46	0.55	0.0	0.0	5484.75	5329.43	1.0
σ	0.10	0.00	0.10	0.11	0.0	0.0	6468.05	5980.43	1.0
nu	1.00	0.02	0.97	1.03	0.0	0.0	6641.91	5786.55	1.0

Image 10: Model 2's (Student-T) Summary Statistics



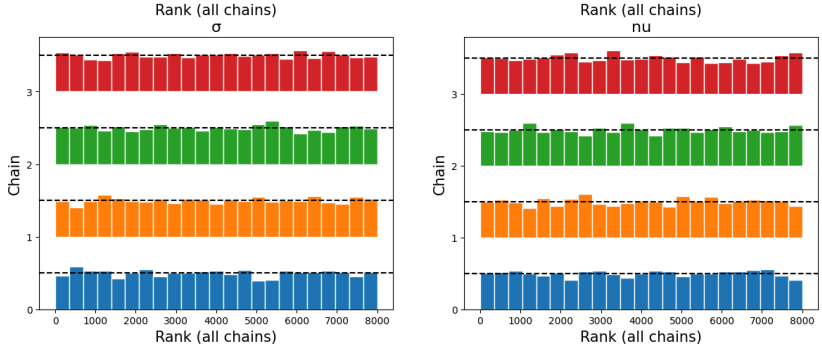


Image 11: Model 2's Rank Plots.

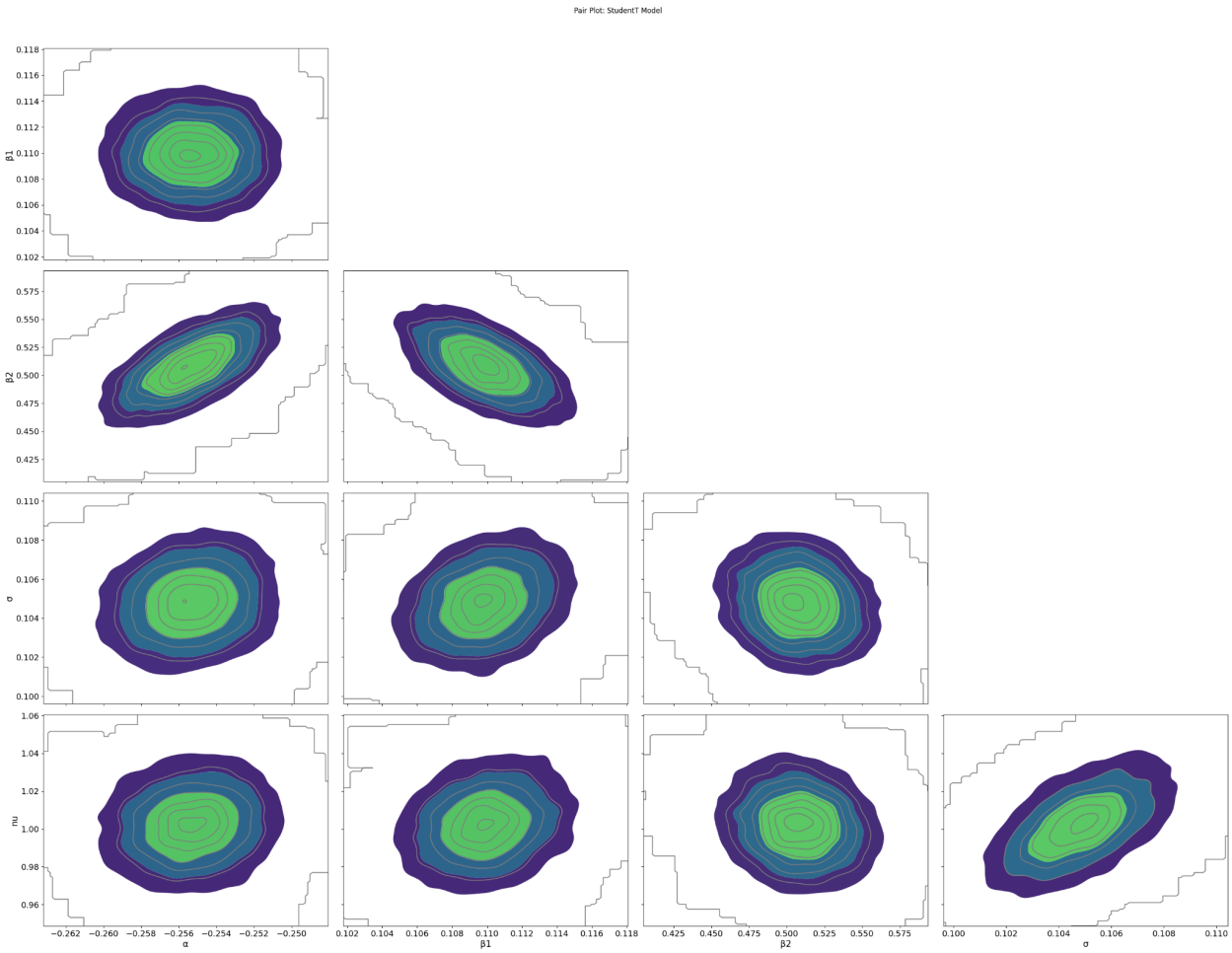


Image 12: Model 2's Pair Plots.

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk
α	−0.25	0.00	−0.25	−0.24	0.0	0.0	3778.94
β1	0.14	0.00	0.14	0.15	0.0	0.0	3798.32
β2	0.17	0.00	0.16	0.17	0.0	0.0	6118.98
is_outlier[0]	0.01	0.11	0.00	0.00	0.0	0.0	5613.42
is_outlier[1]	0.01	0.12	0.00	0.00	0.0	0.0	5210.22
...	...	...	...	...	...	...	...
is_outlier[14198]	0.05	0.21	0.00	0.00	0.0	0.0	5843.84
is_outlier[14199]	1.00	0.00	1.00	1.00	0.0	0.0	8000.00
sigma_in	0.16	0.00	0.16	0.16	0.0	0.0	2678.36
sigma_out	2.47	0.04	2.38	2.55	0.0	0.0	4946.23
p	0.15	0.00	0.14	0.15	0.0	0.0	3732.30

	ess_tail	r_hat
α	5160.39	1.0
β1	5203.99	1.0
β2	5510.17	1.0
is_outlier[0]	5613.42	1.0
is_outlier[1]	5210.22	1.0
...	...	...
is_outlier[14198]	5843.84	1.0
is_outlier[14199]	8000.00	NaN
sigma_in	4558.09	1.0
sigma_out	5671.03	1.0
p	5075.04	1.0

Image 13: Model 3's (Outlier Classifier) Summary Statistics.

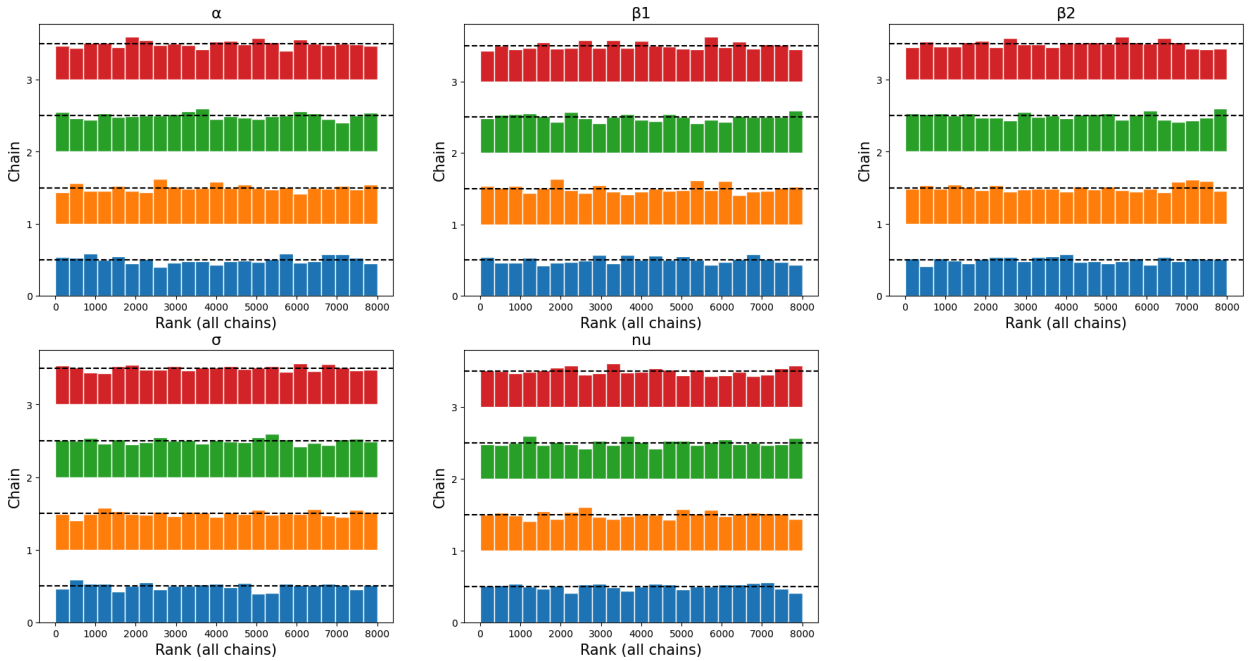


Image 14: Model 3's Rank Plots.

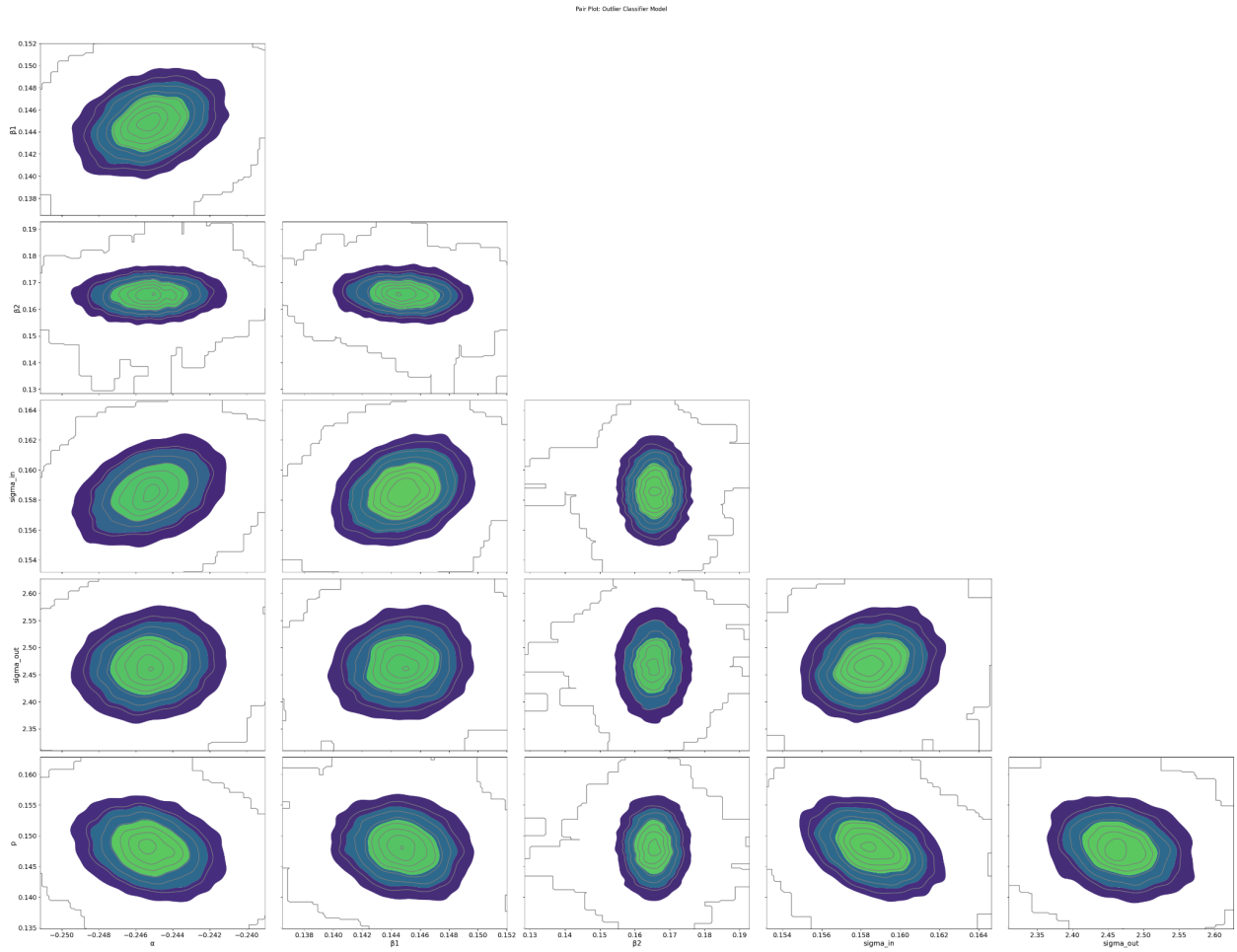


Image 15: Model 3's Pair Plots.